

Technical Requirements Document (TRD)

New Sai Car Bazar — Used Car Dealership Platform

Document Version: 1.0

Date: 27 September 2026

Project Type: Full-stack, production-ready used-car dealership platform

Primary Location: Barabanki, Uttar Pradesh, India

Development Approach: AI-assisted / Vibe Coding

Target Deployment: GitHub + Vercel + Supabase

1. Project Overview

New Sai Car Bazar requires a modern, secure, responsive used-car dealership website based on the provided visual references.

The system must not be a static marketing website. It must be a **fully functional inventory management platform** where the authorized administrator can:

- Add vehicles
- Edit vehicles
- Upload/manage vehicle images
- Mark vehicles as featured/new arrival/hot deal
- Mark vehicles as sold
- Manage enquiries
- Manage finance enquiries
- Manage Sell My Car requests
- Manage vehicle inventory through an admin dashboard

Customers must be able to browse inventory without creating an account.

The platform will use **Supabase** for authentication, PostgreSQL database, storage, and row-level security.

2. Business Information

FieldRequirement Business NameNew Sai Car Bazar LocationBarabanki, Uttar Pradesh, India

Inventory SizeLess than 30 vehicles initially Inventory ManagementManual + CSV bulk import

Admin UsersAdmin only Customer LoginOptional Customer EnquiryNo login required

PaymentsNot required WhatsAppYes Finance CalculatorYes LogoExisting logo will be provided

DomainNot purchased yet GitHubNew repository BackendSupabase DeploymentVercel

Contact Information

Current requirements indicate:

- Phone: 8858982362
- WhatsApp: 8172946630
- Email: mishraprashant2808@gmail.com

Implementation note: the phone/WhatsApp numbers should be confirmed before production deployment because the supplied screenshot and later requirement contain different numbers.

3. Project Objectives

The system shall:

1. Provide a premium used-car dealership website.
1. Closely reproduce the visual style of the supplied screenshots.
1. Provide a responsive experience across desktop, tablet and mobile.
1. Allow the administrator to manage the complete vehicle inventory.
1. Store inventory in Supabase PostgreSQL.
1. Store vehicle images in Supabase Storage.
1. Provide secure admin authentication.
1. Protect database operations using Row Level Security.
1. Validate all user input on both client and server.
1. Provide enquiry and lead management.
1. Provide an EMI/finance calculator.
1. Provide WhatsApp and phone contact actions.
1. Support SEO-friendly vehicle pages.
1. Be deployable through GitHub → Vercel.
1. Be maintainable and extensible for future functionality.

4. Scope

4.1 In Scope

Public Website

- Homepage
- Header
- Navigation
- Search
- Browse by Make
- Featured Listings
- Inventory page
- Vehicle detail pages
- Vehicle filters
- Vehicle sorting
- Wishlist
- Finance calculator
- Finance enquiry
- Sell My Car
- Contact page
- About Us
- Footer
- WhatsApp integration
- Click-to-call
- Responsive design

Admin

- Admin authentication
- Dashboard
- Inventory CRUD
- Vehicle image management
- Featured vehicles
- New arrivals
- Hot deals
- Sold/archived vehicles
- CSV import
- Enquiry management
- Finance enquiry management
- Sell My Car enquiry management
- Audit logging
- Basic site settings

Backend

- Supabase PostgreSQL
- Supabase Auth
- Supabase Storage
- RLS
- Server-side API/actions
- Validation
- Authorization

5. Out of Scope for Initial Version

The following should **not** be implemented unless specifically requested later:

- Online vehicle payment
- Online booking payment
- Loan approval
- Banking integration
- Automated loan disbursement
- Full CRM
- Multi-dealer marketplace
- Dealer-to-dealer inventory
- Real-time auction
- Insurance purchase
- Vehicle registration API
- Automated valuation using third-party APIs

The architecture should remain extensible enough to support these in future.

6. Technology Stack

Frontend

Next.js

TypeScript

React

Tailwind CSS

shadcn/ui

Backend

Next.js Server Actions / Route Handlers

Supabase

PostgreSQL

Authentication

Supabase Auth

Storage

Supabase Storage

Validation

Zod

React Hook Form

Source Control

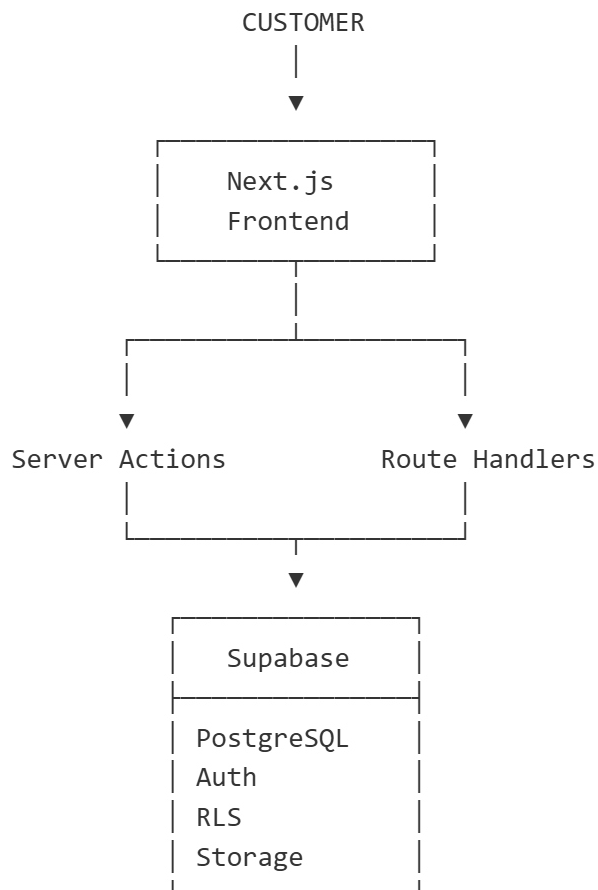
Git

GitHub

Deployment

Vercel

7. High-Level Architecture



8. UI/UX Requirements

The provided screenshots shall be treated as the primary visual reference.

8.1 Visual Direction

The website should use:

- Dark navy backgrounds
- Red primary CTA
- White typography
- Large automotive photography
- Bold headings
- Rounded cards
- Spacious sections
- Premium dealership appearance
- Clear hierarchy
- Strong CTA buttons

Approximate color system:

Primary Red: #ED0000

Dark Navy: #101827

Secondary Navy: #1C293A

White: #FFFFFF

Light Background: #F7F8FA

Final colors should be extracted/refined during implementation based on the supplied design.

9. Public Website Requirements

9.1 Header

Header shall contain:

- Business logo/name
- Search box
- Search button
- Account
- Wishlist indicator
- Navigation

Navigation:

All Vehicles

Home

Inventory

Browse by Make

Finance

Sell My Car

About Us

Contact

Mobile navigation must collapse into a responsive menu.

10. Homepage

Homepage sections:

Hero

Content direction:

QUALITY ASSURED – CERTIFIED PRE-OWNED

FIND YOUR PERFECT
USED CAR

Browse thoroughly inspected used vehicles.
Transparent pricing, full vehicle history,
and flexible finance options.

[Browse Inventory]
[Finance Calculator]

Hero background shall support an automotive image.

10.1 Browse By Make

Makes shall be database-driven.

Example:

Toyota

Ford

Honda

BMW

Mercedes

Audi

Nissan

Jeep

Chevrolet

Volkswagen

Admin must be able to manage makes/models.

10.2 Featured Listings

The homepage shall display vehicles marked:

`is_featured = true`

Cards should support badges:

FEATURED

NEW ARRIVAL

HOT DEAL

The cards shall be database-driven.

11. Inventory Requirements

Route:

`/inventory`

Features:

- Vehicle search
- Make filter
- Model filter
- Price range
- Year range
- Fuel type
- Transmission
- Body type
- Mileage
- Sorting
- Pagination if required
- Vehicle cards
- Vehicle status

Sorting options:

Newest

Price: Low to High

Price: High to Low

Mileage: Low to High

12. Vehicle Detail Page

Route:

/inventory/[slug]

Example:

/inventory/toyota-fortuner-2024

Page shall contain:

- Vehicle image gallery
- Primary image
- Thumbnail gallery
- Vehicle name
- Variant
- Price
- Year
- Mileage
- Fuel
- Transmission
- Body type
- Description
- Features
- Specifications
- Registration information where appropriate
- Vehicle history information where available
- Enquiry CTA
- Call CTA
- WhatsApp CTA
- Finance CTA
- Wishlist CTA

13. Vehicle Status

Vehicle status values:

DRAFT

AVAILABLE

RESERVED

SOLD

ARCHIVED

Only AVAILABLE vehicles shall normally appear in public inventory.

Sold vehicles may optionally have a dedicated "Sold" state/page depending on future SEO requirements.

14. Inventory Data Model

Vehicles Table

vehicles

id
stock_number
slug
make_id
model_id
variant
year
price
mileage
fuel_type
transmission
body_type
color
registration_year
registration_number
vin
description
status
is_featured
is_new_arrival
is_hot_deal
created_at
updated_at
created_by
updated_by

Data Requirements

- stock_number must be unique.
- slug must be unique.
- price must be non-negative.
- mileage must be non-negative.
- year must be a valid vehicle year.
- Required fields must be validated.
- Status must use an approved enum/value set.

15. Makes and Models

Makes

makes

id
name
slug
logo_url
is_active
created_at
updated_at

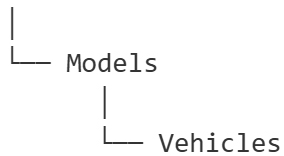
Models

models

id
make_id
name
slug
is_active
created_at
updated_at

Relationship:

Make



16. Vehicle Images

Vehicle Images Table

vehicle_images

id
vehicle_id
storage_path
image_url
alt_text
sort_order
is_primary
created_at

Requirements:

- Multiple images per vehicle
- Primary image
- Reordering
- Delete image
- Upload validation
- File size restrictions
- Allowed image formats
- Image optimization

Recommended formats:

JPG

JPEG

PNG

WEBP

17. Admin Authentication

Admin route:

`/admin/login`

Admin login shall use Supabase Auth.

No public admin signup.

Admin accounts must be created/managed through controlled administrative processes.

18. Admin Dashboard

Route:

`/admin`

Dashboard metrics:

Total Vehicles

Available

Reserved

Sold

Featured

New Arrivals

Hot Deals

New Enquiries

Additional dashboard sections:

- Recent vehicles
- Recent enquiries
- Recent finance requests
- Recent Sell My Car requests

19. Admin Inventory Management

Route:

`/admin/inventory`

Features:

- View vehicles
- Search
- Filter
- Add
- Edit
- Duplicate
- Archive
- Mark sold
- Mark available
- Feature
- Remove featured
- Delete where permitted
- CSV import

20. Add/Edit Vehicle

Route:

`/admin/inventory/new`

`/admin/inventory/[id]`

Form sections:

Basic Information

- Make
- Model
- Variant
- Year
- Price
- Mileage

Specifications

- Fuel
- Transmission
- Body type
- Color

Registration

- Registration year
- Registration number
- VIN, if applicable

Description

Rich text or controlled text editor.

Features

Example:

ABS

Air Conditioning

Airbags

Sunroof

Parking Camera

Cruise Control

Alloy Wheels

Power Windows

Central Locking

Images

Multi-image upload.

Marketing

Featured

New Arrival

Hot Deal

Status

Draft

Available

Reserved

Sold

Archived

21. CSV Import

Admin shall be able to upload a CSV file containing vehicle information.

Requirements:

1. Upload CSV.
1. Validate column headers.
1. Validate every row.
1. Show validation errors before import.
1. Allow user to cancel.
1. Import valid records.
1. Report failed records.
1. Prevent duplicate stock numbers.
1. Generate an import summary.

Example:

Total Rows: 20

Valid: 18

Invalid: 2

Imported: 18

Failed: 2

22. Enquiry System

Customers do not need an account to submit enquiries.

Lead Table

leads

id

vehicle_id

name

phone

email

message

source

status

created_at

updated_at

assigned_to

Status:

NEW

CONTACTED

FOLLOW_UP

CONVERTED

CLOSED

23. Enquiry Form

Required:

Name

Phone

Optional:

Email

Message

Vehicle information should automatically be associated with the enquiry.

Security requirements:

- Server-side validation
- Rate limiting
- CAPTCHA/anti-spam
- Input sanitization
- No direct database insertion from untrusted client code

24. WhatsApp

Vehicle detail page shall provide a WhatsApp CTA.

The message should automatically include vehicle context.

Example:

Hello, I am interested in the Toyota Fortuner 2024
listed on New Sai Car Bazar.

The WhatsApp number shall be configurable through site settings rather than hard-coded throughout the application.

25. Click-to-Call

A call CTA shall use the configured dealership phone number.

Mobile:

tel:+91XXXXXXXXXX

Desktop users may be shown the phone number.

26. Wishlist

Wishlist shall support:

Guest

Use browser/local storage.

Authenticated Customer

Store wishlist in database.

Wishlist Table

wishlists

id

user_id

vehicle_id

created_at

Unique constraint:

user_id + vehicle_id

27. Customer Authentication

Customer authentication is optional.

The public website must remain usable without login.

Customers must be able to:

- Browse vehicles
- Search
- Filter
- View details
- Enquire
- Call
- WhatsApp
- Use finance calculator

without authentication.

Account functionality may be used for:

- Persistent wishlist
- Future customer features

28. Finance Calculator

Route:

/finance

Inputs:

Vehicle Price

Down Payment

Interest Rate

Loan Tenure

Outputs:

Loan Amount

Monthly EMI

Total Interest

Total Payable

EMI formula:

$$EMI = P \times r \times (1+r)^n / ((1+r)^n - 1)$$

Where:

P = principal

r = monthly interest rate

n = number of monthly payments

If the interest rate is zero, the calculator must handle that case separately.

29. Vehicle-Specific Finance

From vehicle detail:

Finance This Car

shall open the calculator with:

Vehicle Price = selected vehicle price

pre-populated.

30. Finance Enquiry

Finance enquiry fields:

Name

Phone

Email

Vehicle

Vehicle Price

Loan Amount

Down Payment

Preferred Tenure

Finance Leads Table

finance_leads

id

vehicle_id

name

phone

email

vehicle_price

loan_amount

down_payment

interest_rate

tenure

message

status

created_at

updated_at

Status:

NEW

CONTACTED

FOLLOW_UP

CLOSED

31. Sell My Car

Route:

/sell-my-car

Fields:

Make

Model

Year

Mileage

Registration

Expected Price

Name

Phone

Email

Description

Photos

Sell Car Requests

sell_car_requests

id

make

model

year

mileage

registration

expected_price

name

phone

email

description

status

created_at

updated_at

Optional vehicle images can be stored in a dedicated Supabase Storage path.

32. About Us

The About page shall reproduce the visual direction of the provided reference.

Content sections may include:

- Why Buy From New Sai Car Bazar
- Certified inspection
- Vehicle history
- Flexible finance
- Customer support
- Business information

All claims about inspections, warranties, reports, financing etc. should only be displayed if actually provided by the dealership.

33. Contact Page

Contact page should contain:

- Business name
- Location
- Phone
- WhatsApp
- Email
- Contact form
- Map/location section if desired

Contact form submissions should enter the lead system.

34. Footer

Footer shall contain:

About Us

Short business description.

Quick Links

About Us

Contact Us

Sell My Car

Finance Options

More Info

Buying Guide

Car Value Estimator

FAQ

Terms & Conditions

New Arrivals

Email subscription UI may be implemented as a future-ready component.

35. Database Security

Supabase Row Level Security is mandatory.

Public

Public users should only be able to read data explicitly intended for public consumption.

Example:

`AVAILABLE vehicles → public read`

Private lead information must never be publicly readable.

36. Admin Authorization

Admin operations must be protected at multiple levels:

User Authentication

↓

Role Verification

↓

Server Authorization

↓

Supabase RLS

↓

Database

Frontend-only authorization is prohibited.

37. Sensitive Data

The following must never be publicly exposed:

- Customer phone numbers from leads
- Customer email addresses
- Finance information
- Sell My Car submissions
- Admin data
- Service-role credentials
- Database credentials
- Private API keys

38. Environment Variables

Required environment variables will be managed through `.env.local` and Vercel environment settings.

Example:

```
NEXT_PUBLIC_SUPABASE_URL=
```

```
NEXT_PUBLIC_SUPABASE_ANON_KEY=
```

```
SUPABASE_SERVICE_ROLE_KEY=
```

The service-role key must never be exposed to browser/client code.

`.env.local` must be excluded from Git.

39. Validation Requirements

Validation must happen on both:

Client

For user experience.

Server

For security.

Zod schemas should be centralized:

```
lib/validation/
```

```
├─ vehicle.ts
```

```
├─ lead.ts
```

```
├─ finance.ts
```

```
├─ sell-car.ts
```

```
└─ auth.ts
```

40. File Upload Security

Uploaded files must be checked for:

- MIME type
- File extension
- File size
- Upload path
- User authorization

Only permitted image formats should be accepted.

Server-side checks are mandatory.

41. Rate Limiting

Rate limiting should apply to public forms and sensitive endpoints:

Contact

Vehicle Enquiry

Finance Enquiry

Sell My Car

Authentication

This helps prevent:

- Spam
- Brute-force login attempts
- Automated form abuse
- Excessive API requests

42. CAPTCHA

Public forms should use a CAPTCHA/anti-bot mechanism such as Cloudflare Turnstile or an equivalent service.

CAPTCHA verification must happen server-side.

43. Security Headers

Production application should implement appropriate HTTP security headers, including where compatible:

Content-Security-Policy

X-Content-Type-Options

Referrer-Policy

Permissions-Policy

Strict-Transport-Security

Frame protections

Configuration must not break required third-party services.

44. Audit Logs

Admin changes should be logged.

Audit Logs

audit_logs

id

user_id

action

entity_type

entity_id

old_data

new_data

ip_address

user_agent

created_at

Examples:

VEHICLE_CREATED
VEHICLE_UPDATED
VEHICLE_ARCHIVED
VEHICLE_STATUS_CHANGED
IMAGE_UPLOADED
IMAGE_DELETED
LEAD_STATUS_CHANGED

Sensitive values should not be unnecessarily logged.

45. SEO Requirements

Each vehicle page should have:

- Unique title
- Meta description
- Canonical URL
- Open Graph metadata
- Social image
- Semantic HTML
- Structured data where appropriate
- SEO-friendly slug

Example:

/inventory/toyota-fortuner-2024

not:

/car?id=1827

46. Performance Requirements

Target:

- Fast initial page load
- Optimized images
- Lazy-loaded images where appropriate
- Responsive image sizes
- Minimal client-side JavaScript
- Server-side rendering where appropriate
- Database queries limited to required fields
- Indexed searchable database fields

Images must not be loaded at their original massive resolution when a smaller optimized version is sufficient.

47. Responsive Requirements

The website must work on:

Mobile

Tablet

Laptop

Desktop

Large Desktop

Critical pages:

- Homepage
- Inventory
- Vehicle detail
- Finance
- Contact
- Admin dashboard

must all be responsive.

48. Accessibility

Target:

- Semantic HTML
- Keyboard navigation
- Visible focus states
- Accessible form labels
- Proper button semantics
- Alt text for vehicle images
- Sufficient color contrast
- Accessible modal/dialog behavior

49. Error Handling

User-facing errors should be friendly.

Example:

Something went wrong.

Please try again.

Technical details should not be exposed to users.

Server logs may contain technical information where appropriate, without exposing secrets or sensitive customer data.

50. Logging

Application logging shall support:

- Errors
- Authentication failures
- Important admin operations
- Server errors
- Import failures

Logs must not contain:

Passwords

API keys

Service-role keys

Full payment information

Unnecessary PII

51. GitHub Requirements

Repository:

`new-sai-car-bazar`

Recommended branches:

```
main
development
feature/*
```

Pull requests should be used for significant changes once development expands.

52. Git Commit Standards

Recommended:

```
feat: add vehicle inventory
feat: add finance calculator
fix: correct vehicle filtering
security: update RLS policies
refactor: improve vehicle card
docs: update setup instructions
```

53. Environment Separation

Minimum:

Development

Production

Recommended later:

Development

Preview

Production

Vercel preview deployments can be used for testing changes before production.

54. Supabase Storage Structure

Recommended:

```
vehicle-images/  
    {vehicle-id}/  
        01.webp  
        02.webp  
        03.webp
```

```
sell-car/  
  {request-id}/  
    01.webp  
    02.webp
```

Storage policies must prevent unauthorized uploads/deletes.

55. Database Indexes

Indexes should be created for commonly queried fields such as:

```
vehicles.status
vehicles.make_id
vehicles.model_id
vehicles.price
vehicles.year
vehicles.created_at
vehicles.slug
vehicles.stock_number
```

Composite indexes should be added only where actual query patterns justify them.

56. Testing Requirements

Functional

Test:

- Admin login
- Add vehicle
- Edit vehicle
- Delete/archive vehicle
- Upload images
- Mark featured
- Mark sold
- Search
- Filters
- Sorting
- Enquiry
- WhatsApp
- Call
- Finance calculator
- Finance enquiry
- Sell My Car
- CSV import
- Wishlist

57. Security Testing

Test:

- Unauthorized admin access
- Direct database access
- RLS policies
- Invalid form data
- Malicious input
- File upload abuse
- Rate limiting
- CAPTCHA bypass attempts
- Session handling
- Role escalation
- Public access to private leads

58. Acceptance Criteria

The project will be considered MVP-complete when:

Public

- Homepage matches supplied design direction.
- Responsive header works.
- Inventory loads from Supabase.
- Search works.
- Filters work.
- Vehicle detail pages work.
- Images load from storage.
- Enquiry works without login.
- Call button works.
- WhatsApp works.
- Finance calculator works.
- Finance enquiry works.
- Sell My Car works.
- Wishlist works.
- About/Contact pages work.

Admin

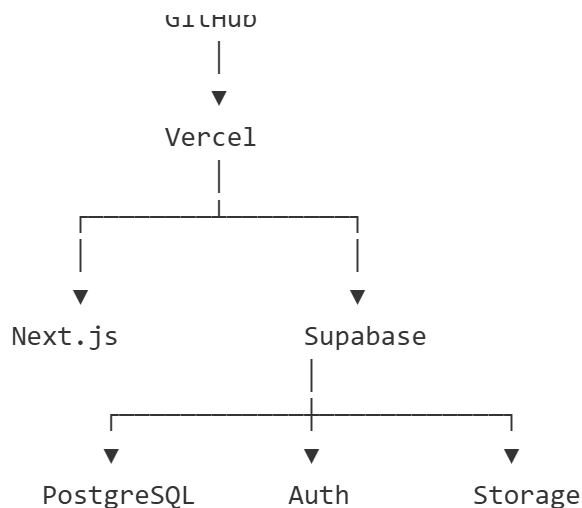
- Admin login works.
- Unauthorized users cannot access admin.
- Vehicle CRUD works.
- Images can be uploaded.
- Images can be reordered.
- Featured status works.
- New Arrival works.
- Hot Deal works.
- Sold status works.
- CSV import works.
- Leads are visible.
- Lead status can be updated.
- Finance leads are visible.
- Sell My Car leads are visible.
- Audit logging works.

Security

- RLS enabled.
- Private data protected.
- Server validation implemented.
- Client validation implemented.
- Secrets not committed.
- Service-role key not exposed.
- Upload validation implemented.
- Rate limiting implemented.
- CAPTCHA implemented.
- Production security headers configured.

59. Deployment Requirements

Final architecture:



Deployment sequence:

1. Create GitHub repository
2. Create Supabase project
3. Configure database
4. Configure RLS
5. Configure Storage
6. Configure Auth
7. Build application
8. Push to GitHub
9. Connect GitHub to Vercel
10. Configure environment variables
11. Deploy preview
12. Test
13. Deploy production
14. Connect custom domain later

60. Future Domain

Since no domain currently exists, initial deployment can use the Vercel-provided URL.

Once the business purchases a domain:

Domain



Vercel



New Sai Car Bazar

SSL should be automatically configured through the deployment platform.

61. Recommended Development Phases

Phase 1 — Project Foundation

Next.js

TypeScript

Tailwind

shadcn

GitHub

Environment setup

--

--

Phase 2 — UI

Header

Navbar

Hero

Browse by Make

Featured Vehicles

Trust Section

Footer

Responsive design

Phase 3 — Supabase

Database

Auth

Storage

RLS

Migrations

Seed data

Phase 4 — Inventory

CRUD

Images

Search

Filters

Sorting

Vehicle details

Phase 5 — Admin

Dashboard

Inventory manager

CSV import

Lead management

Audit logs

Phase 6 — Customer Features

Enquiries

Wishlist

Finance

Sell My Car

WhatsApp

Call

Phase 7 — Security

RLS

Validation

Rate limiting

CAPTCHA

Headers

Upload security

Authorization

Phase 8 — Testing

Functional

Functional
Security
Responsive
Performance
SEO
Accessibility

Phase 9 — Deployment

GitHub
Vercel
Supabase
Production

62. AI/Vibe-Coding Development Rules

Because this project will be built with AI assistance, the coding agent must follow these rules:

Rule 1

Do not build everything in one giant generation.

Build incrementally.

Rule 2

Do not hard-code inventory.

Vehicle data must come from Supabase.

Rule 3

Do not trust client-side authorization.

All privileged operations require server-side authorization and database policies.

Rule 4

Do not expose secrets.

Never place:

`SUPABASE_SERVICE_ROLE_KEY`

database credentials

private API keys

in client code.

Rule 5

Do not destroy existing functionality while implementing new features.

Rule 6

After each major feature:

Build

Lint

Type-check

Test

Review

Rule 7

Database changes must use versioned migrations.

Rule 8

Do not use fake/mock inventory in production.

Seed data may be used only for development.

63. Definition of Done

A feature is not considered complete merely because its UI exists

A feature is not considered complete merely because its UI exists.

For example:

"Add Vehicle" is complete only when:

UI

↓

Validation

↓

Authorization

↓

Server operation

↓

Database

↓

Image Storage

↓

Public Inventory

↓

Error handling

↓

Testing

all work correctly.

The same principle applies to every major feature.

64. Initial MVP

The first production milestone shall include:

- ✓ Premium responsive homepage
- ✓ Inventory
- ✓ Vehicle detail
- ✓ Search
- ✓ Filters
- ✓ Admin authentication
- ✓ Admin dashboard
- ✓ Vehicle CRUD
- ✓ Image upload
- ✓ Supabase database
- ✓ Supabase storage
- ✓ RLS
- ✓ Server validation
- ✓ Enquiry system
- ✓ WhatsApp
- ✓ Call
- ✓ Finance calculator
- ✓ Finance enquiry
- ✓ Sell My Car
- ✓ Wishlist
- ✓ CSV inventory import
- ✓ Audit logs
- ✓ GitHub
- ✓ Vercel deployment

65. Pending Business Inputs

The following remain to be finalized before production:

ItemStatus LogoAwaiting upload Exact showroom addressPending Phone number confirmationPending WhatsApp number confirmationProvided: 8172946630 Default EMI interest ratePending DomainNot purchased Supabase projectTo be created GitHub repositoryTo be created

CC BY-NC-SA 4.0